

SCIENCE

1. Passage anchor

> “I still remember walking into the classroom feeling slightly lost, because **science** had never been my strongest subject.”

Later:

> “She teaches senior physics classes, helps with the school’s **science club**, and mentors students who want to go into **science-related fields**.”

Passage meaning

In the first sentence:

science = the school subject area dealing with the natural world, often including physics, chemistry, biology, etc.

科學；理科；科學科目

Here, the speaker means that school science was not their strongest academic area.

Practice

1. **Science** was never my easiest subject at school.

科學 / 理科從來都不是我在學校最拿手的科目。

2. She helped me become much more confident in **science**.

她幫助我在**科學科目**上變得更有自信。

Core meanings of SCIENCE

2. science — systematic study of the natural world

= the systematic study of the physical and natural world through observation, evidence, and experiment

科學

Examples

1. **Science** helps us understand how the natural world works.

科學幫助我們了解自然世界如何運作。

2. Modern **science** depends on evidence and testing.

現代**科學**依靠證據與驗證。

This is the central broad meaning.

3. science — school subject

= science taught as an academic subject, often covering several scientific disciplines
科學科；理科

This is the exact passage sense.

Examples

1. **Science** was my favourite subject in primary school.
科學科是我小學時最喜歡的科目。
2. We had a **science lesson** this morning.
我們今天早上上了一堂**科學課**。

4. study science

= formally learn scientific subjects
學科學；修讀理科

Examples

1. She wants to **study science** at university.
她想在大學**修讀理科**。
2. Students who **study science** often take laboratory classes.
修讀科學科目的學生通常會上實驗課。

Branch / field of knowledge

5. a science

= a particular organized field of scientific knowledge
一門科學；一個科學領域

Examples

1. Biology is a **science**.

生物學是一門**科學**。

2. Astronomy developed into a modern **science** over time.

天文學逐漸發展成一門現代**科學**。

Here, **science** is countable.

6. the sciences

= scientific disciplines considered collectively

各門科學；自然科學

Examples

1. Mathematics is important in many of **the sciences**.

數學在許多**科學領域**中都很重要。

2. The university is strong in **the sciences**.

這所大學在**理科領域**很強。

Natural science

7. natural science

= science concerned with the physical and biological world

自然科學

Examples

1. Physics, chemistry, and biology are major **natural sciences**.

物理、化學和生物都是主要的**自然科學**。

2. She has always been interested in **natural science**.

她一直對**自然科學**很有興趣。

Social science

8. social science

= systematic study of human society and social behavior

****社會科學****

Examples

1. Economics is generally considered a ****social science****.

經濟學通常被視為一門****社會科學****。

2. She studies ****social science**** at university.

她在大學修讀****社會科學****。

This is important because ****science**** is not limited only to physics, chemistry, and biology.

Applied / technical branches

9. computer science

= academic field dealing with computation, algorithms, software, data, and related theory

****電腦科學；計算機科學****

Examples

1. He studies ****computer science****.

他修讀****電腦科學****。

2. ****Computer science**** includes both theory and practical applications.

****電腦科學****同時包括理論和實際應用。

10. data science

= field using statistics, computing, and analytical methods to extract knowledge from data

****資料科學；數據科學****

Examples

1. She works in ****data science****.

她從事****資料科學****工作。

2. ****Data science**** combines programming, statistics, and analysis.

****資料科學****結合理論設計、統計和分析。

11. environmental science

= scientific study of the environment and environmental systems

****環境科學****

Examples

1. She studied ****environmental science**** at university.
她在大學修讀****環境科學****。
2. ****Environmental science**** examines interactions between natural systems and human activity.
****環境科學****研究自然系統與人類活動之間的互動。

Scientific knowledge / method

12. science — scientific knowledge

= knowledge obtained through scientific investigation

****科學知識****

Examples

1. The treatment is supported by ****science****.
這種治療有****科學證據****支持。
2. Good policy should take account of the ****science****.
好的政策應該考慮相關的****科學證據 / 科學研究****。

Here, ****science**** can mean the body of scientific findings, not the school subject.

13. the science behind something

= scientific principles or evidence that explain why something works

****.....背後的科學原理****

Examples

1. The programme explains the ****science behind sleep****.
這個節目解釋****睡眠背後的科學原理****。

2. I wanted to understand the **science** behind the experiment.

我了解**這個實驗背後的科學原理**。

Systematic skill / technique

14. science — systematic body of knowledge or technique

= a subject or activity treated as something that can be studied systematically

有系統的方法；學問

Examples

1. There is a **science** to good teaching.

良好的教學其實也有一套**系統性方法**。

2. Negotiation is partly an art and partly a **science**.

談判既是一門藝術，也是一門**講求方法與規律的學問**。

This is an extended figurative sense.

15. an art and a science

= something requiring both creativity/judgment and systematic knowledge/method

既是藝術，也是科學；兼具技巧與系統方法

Examples

1. Good leadership is both an art and a **science**.

良好的領導既是一門藝術，也是一門**科學**。

2. Teaching can be seen as both an art and a **science**.

教學可以被視為既是藝術，也是**有系統的方法**。

Institutional / educational uses

16. science club

= school or community club focused on science activities

科學學會；科學社

This appears directly in the passage.

Examples

1. She helps run the school **science club**.
她協助管理學校的**科學學會**。
2. The **science club** organizes experiments and competitions.
科學社會舉辦實驗活動和比賽。

17. science-related field

= academic or professional area connected with science
與科學相關的領域；理科相關領域

This also appears directly in the passage.

Examples

1. Many students want to enter **science-related fields**.
很多學生想進入**科學相關領域**。
2. Engineering is closely connected with several **science-related fields**.
工程學和多個**理科相關領域**有密切關係。

18. science education

= teaching and learning of science
科學教育

Examples

1. She has worked in **science education** for many years.
她從事**科學教育**多年。
2. Good **science education** encourages curiosity and evidence-based thinking.
良好的**科學教育**鼓勵好奇心和以證據為本的思考。

Related forms

19. scientist

= person who studies or works professionally in science

****科學家****

Examples

1. She hopes to become a ****scientist****.
她希望成為一名****科學家****。
2. The ****scientists**** tested the new theory.
****科學家們****測試了新理論。

20. scientific

= relating to science or using scientific methods

****科學的；科學性的****

Examples

1. The claim needs ****scientific evidence****.
這項說法需要****科學證據****。
2. The researchers used a ****scientific method****.
研究人員使用了****科學方法****。

21. scientifically

= in a way based on science or scientific methods

****科學地；從科學角度****

Examples

1. The idea has not been ****scientifically tested****.
這個想法尚未經過****科學驗證****。
2. The phenomenon can be explained ****scientifically****.
這個現象可以從****科學角度****解釋。

22. unscientific

= not based on proper scientific evidence or method

****不科學的；缺乏科學依據的****

Examples

1. The claim is ****unscientific****.

這項說法****缺乏科學依據****。

2. The method was criticized as ****unscientific****.

這個方法被批評為****不科學****。

Countability

23. science can be uncountable or countable

Uncountable

when referring to science generally:

> ****Science**** changes as new evidence appears.

> ****科學****會隨著新證據出現而發展。

Countable

when referring to a particular discipline:

> Biology is a ****science****.

> 生物學是一門****科學****。

Plural

when referring to several disciplines:

> the natural ****sciences****

> ****自然科學各領域****

This distinction is useful for MCQs.

Grammar / capitalization

24. science usually takes a lowercase s

Normally:

- * science
- * science teacher
- * science club
- * science subject

Capitalize it when it begins a sentence or is part of an official title.

Key contrasts

25. science vs physics

****science****

= broad field covering many systematic studies of the natural world and beyond

****科學****

****physics****

= specific branch studying matter, energy, forces, motion, etc.

****物理學****

Compare:

> Physics is a branch of ****science****.

> 物理學是****科學****的一個分支。

So in the passage:

* ****science**** = broad subject area

* ****physics**** = Ms. Lau's specific subject

26. science vs subject

****science****

= particular area of knowledge

****subject****

= any area studied at school or university

****科目****

So:

> Science is a **subject**.

but not every subject is science.

27. science vs knowledge

knowledge

= information and understanding someone possesses

知識

science

= organized method/system for developing and testing knowledge about the world

科學

Science produces knowledge, but the two are not identical.

28. science vs research

science

= broader field and method of inquiry

research

= systematic investigation undertaken to answer a particular question or produce knowledge

研究

Compare:

> She works in **science**.

> 她從事科學相關工作。

> She conducts **research** on light.

> 她進行關於光的研究。

29. science vs technology

science

= primarily seeks to understand how things work

科學

****technology****

= uses knowledge and tools for practical purposes

****科技 ; 技術****

Compare:

> Science helps explain electricity.

> Technology uses electrical knowledge to build devices.

They overlap heavily but are not interchangeable.

30. science vs engineering

****science****

= seeks explanations and understanding

****engineering****

= applies scientific and mathematical knowledge to design and solve practical problems

****工程學****

Physics is a science; engineering often applies physics.

31. science vs scientific

****science****

= noun

****科學****

****scientific****

= adjective

****科學的 ; 以科學方法進行的****

Compare:

> She studies ****science****.

> 她學科學。

> She conducted a ****scientific experiment****.

> 她進行了一個科學實驗。

Meaning map

The central concept of **science** is:

systematic investigation of the world using evidence, observation, and testable methods

From this develop:

General field

science

→ **科學**

School subject

science

→ **科學科 / 理科**

Individual discipline

a science

→ **一門科學**

Collective disciplines

the sciences

→ **各門科學**

Scientific evidence

the science

→ **相關科學研究 / 科學證據**

Explanation

the science behind something

→ **背後的科學原理**

Systematic method

there is a science to it

→ **有一套系統性方法**

And the passage adds educational compounds:

science club

→ **科學學會**

science-related field

→ **科學相關領域**

The passage's first use is specifically the **school-subject** branch.

MCQ-ready master list

Form / sense	MCQ-ready Traditional Chinese meaning
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science — general field	透過觀察、證據與實驗系統研究自然世界的科學
science — school subject	在學校中學習物理、化學、生物等內容的科學科 / 理科
study science	正式學習或修讀科學類科目
a science	一個具系統理論與研究方法的科學領域
the sciences	多個科學學科的總稱；各門科學
natural science	研究自然界、物質世界與生物世界的自然科學
social science	系統研究人類社會與社會行為的社會科學
computer science	研究計算、演算法、軟體等的電腦科學
data science	利用統計、運算與分析方法從資料中獲取知識的資料科學
environmental science	研究環境及其系統的環境科學
science — body of evidence/knowledge	某議題已有的科學研究、證據與知識
the science behind something	解釋某現象如何運作的科學原理
science — systematic method	一套可系統研究、分析和掌握的方法或學問
an art and a science	同時需要創意判斷和系統方法的活動
science club	以科學活動、實驗與學習為主的科學學會 / 科學社
science-related field	與科學知識、研究或應用有關的學術 / 職業領域
science education	科學知識與方法的教學和學習；科學教育
scientist	專業從事科學研究的人；科學家
scientific	與科學有關或按照科學方法進行的
scientifically	以科學方法或從科學角度
unscientific	缺乏科學方法或證據支持的；不科學的

Passage recap

> "...because **science** had never been my strongest subject."

Here:

****science**** = ****science as a school subject / academic area****

Best Traditional Chinese:

* ****科學****

* ****理科****

* ****科學科目****

Natural contextual translation:

****因為理科從來都不是我最擅長的科目。****

Passage MCQ-ready meaning

****science****

= ****the school subject concerned with understanding the natural world through areas such as physics, chemistry, and biology****

****科學科；理科；科學****

The later expressions use the broader field sense:

> ****science club****

> → 科學學會

> ****science-related fields****

> → 科學相關領域

So the passage itself already shows two closely related uses:

****science as a school subject****

→ ****理科 / 科學科****

and

****science as a broader academic/professional domain****

→ ****科學領域****.